



The impact of top management team's innovation orientation on organizational ambidexterity in transitional economies: Evidence from Bosnia and Herzegovina

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ABSTRACT

Organizational ambidexterity plays an essential role in understanding how firms compete and prosper in dynamic environments. However, our understanding of the antecedents and effects of organizational ambidexterity is primarily derived from studies conducted in developed economies, while transitional economies with their unique characteristics and challenges are less well understood. Grounded in upper echelon theory, this paper contends that the innovation orientation of a firm's top management team (TMT) impacts organizational ambidexterity and firm performance. Drawing on a dataset of 192 small and medium-sized companies from Bosnia and Herzegovina, and estimating several OLS regressions, we find evidence that TMT's innovation orientation negatively impacts firm performance in transitional economies. Further, we see that the impact of the TMT's innovation orientation on the balance dimension of organizational ambidexterity is inverted U-shaped. Organizational innovation culture leverages this relation. Intriguingly, we found that in transitional economies, organizational ambidexterity was not correlated with firm performance, but its components exploration and exploitation were, albeit in opposite directions. Overall, our work adds nuanced insights to the antecedents and effects of organizational ambidexterity in transitional economies.

1. Introduction

Organizational ambidexterity, i.e., the proficiency of a firm to succeed in both exploiting given opportunities and exploring new avenues, has emerged as a focal point in scholarly discussions on how firms prosper in dynamic environments (Christensen, 1997; Gupta et al., 2006; Kassotaki, 2022; Tarba et al., 2020; Tushman & O'Reilly, 1996). Within this stream of research, structural ambidexterity plays an important role in understanding why some firms are more successful than others (Fourné et al., 2019; Hansen et al., 2019; O'Reilly & Tushman, 2013). To become structurally ambidextrous, firms need to establish organizational designs which allow them to balance exploitative and explorative activities (Jansen, Tempelaar, et al., 2009; Lawrence & Lorsch, 1967). Balancing exploration and exploitation involves both the separation through specialized organizational units and the simultaneous integration of these units by making use of integration

mechanisms (Andriopoulos & Lewis, 2009; Benner & Tushman, 2003; Gupta et al., 2006; Raisch, 2008; Raisch et al., 2009; Tushman & O'Reilly, 1996). Among those mechanisms, behavioral integration of the top management team (TMT) represents a pivotal antecedent of organizational ambidexterity (Carmeli & Halevi, 2009; Koryak et al., 2018; Lubatkin et al., 2006; Simsek, 2009; Venugopal et al., 2020). Behavioral integration refers to the ability of a TMT to make demanding choices (Hambrick & Mason, 1984) in respect to exploitation and exploration (Lubatkin et al., 2006; Simsek, 2009). Behavioral integration is, for example, grounded in joint decision-making processes, a high-level of information exchange or a commonly shared vision (Andriopoulos & Lewis, 2009; Lubatkin et al., 2006; Venugopal et al., 2020). Overall, the TMT and behavioral integration are important antecedents of organizational ambidexterity.

However, research on organizational ambidexterity predominantly draws from studies conducted in developed economies (Batra et al.,

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2022). Yet, increased levels of volatility, uncertainty and complexity, as prevalent in emerging and transitional economies,¹ introduce a distinctive set of challenges (Ferraris et al., 2022; Shankar & Narang, 2020) in achieving organizational ambidexterity. In contrast to developed economies, emerging and transitional economies show unique characteristics and grapple with different issues such as bold heterogeneity, sociopolitical governance challenges, a less developed and supportive innovation system, resource scarcity, unbranded competition, widespread corruption and deficiencies in infrastructure (Carney et al., 2019; Kostova & Marano, 2018; Round, 2009; Sheth, 2011; Stojčić, 2024; Stojčić, 2021; Stojčić et al., 2024). Further, these economies stand out due to the pervasive impact of ingrained cultural values, hierarchical structures and low-levels of entrepreneurial orientation and innovation capabilities (Bahl et al., 2021; Burgess & Steenkamp, 2006; Dabić et al., 2021; Estrin et al., 2009; Hervas-Oliver et al., 2021; Stojčić, 2024). These idiosyncrasies are particularly evident in the Southeast European (SEE) region (Bahl et al., 2021; Estrin et al., 2009; Kleibrink et al., 2018; Kroll et al., 2017; Pilav – Velic, Jahic, & Krndzija, 2024; Pilav-Velić & Marjanovic, 2016; Radosevic, 2017; Stojčić et al., 2024; Vujanović et al., 2022; Švarc & Dabić, 2021). In SEE, socialist values such as collectivism, shared responsibility, employee involvement, suppression of risk-taking, limited entrepreneurial mindset, and missing capabilities to facilitate learning shape how businesses operate and how leaders manage firms. This institutional setting forms TMT's innovation orientation in a unique way, often favoring cautious and incremental innovation strategies over bold and market-disruptive approaches. Further on, in emerging and transitional economies, “[l]imited innovation capacities make their firms more risk averse to innovation failure than firms in advanced innovation systems” (Stojčić, 2024, p. 2).

Taken together, the unique characteristics of transitional economies go beyond environmental dynamism as important contingency factor explaining and understanding organizational ambidexterity (Jansen, Vera, & Crossan, 2009; Luger et al., 2018; Tamayo-Torres et al., 2017). However, research has largely failed to fully account how these idiosyncrasies shape organizational ambidexterity and relate to firm performance (Batra et al., 2022). One valuable exception is Pilav – Velic et al. (2024) who investigate the moderating role of firm's resilience on the impact of exploitation and exploration on innovation performance in Bosnia and Herzegovina (BiH). Yet, Pilav – Velic et al. (2024) do not reflect upon the antecedents of organizational ambidexterity nor on general firm performance implications. Further taking into account that organizational culture is closely linked to organizational leaders and firm performance (O'Reilly et al., 2014, 2025; Zhang et al., 2023), we propose the following research question:

How does TMT's innovation orientation impact organizational ambidexterity and firm performance in transitional economies, and how is this impact influenced by organizational innovation culture?

Based on upper echelon theory (Carpenter et al., 2004; Hambrick & Mason, 1984; Neely et al., 2020), we investigate the impact of TMT's innovation orientation on organizational ambidexterity and firm performance. Thereby, we refer to the balance dimension of organizational ambidexterity, henceforth organizational ambidexterity (BD) (Cao et al., 2009; Kassotaki, 2022; Venugopal et al., 2020). Data from BiH, an economy which is commonly considered as a transitional economy (Ateljevic et al., 2004; Round, 2009; United Nations, 2024), shows that, from a certain threshold on, TMT's innovation orientation becomes dysfunctional for achieving organizational ambidexterity (BD), meaning that too much of TMT's innovation orientation leads to an imbalance in organizational ambidexterity (BD). Further, we see that the inverted

U-shaped relation between TMT's innovation orientation and organizational ambidexterity (BD) is moderated by organizational innovation culture, whereas strong organizational innovation cultures reinforce the negative impact of TMT's innovation orientation on organizational ambidexterity (BD), but only for firms with a high level of TMT's innovation orientation. Additionally, we find evidence that organizational ambidexterity (BD) is not correlated with firm performance in transitional economies, thereby challenging the notion that it is of special importance for firms in emerging and transitional economies to become ambidextrous (Batra et al., 2022; Hsu et al., 2013; Hu et al., 2023). Overall, our findings contribute to the scarce literature on organizational ambidexterity in transitional economies, by showing that the antecedents and effects of organizational ambidexterity in these economies are unique and differ from those in developed economies, and by identifying a set of practical implications for the management of firms in transitional economies on their path towards organizational ambidexterity and firm performance.

2. Theoretical background

2.1. Organizational ambidexterity

Organizational ambidexterity is commonly understood as the ability of firms to simultaneously pursue exploration and exploitation rather than focusing one-sidedly on one of these activities (Andriopoulos & Lewis, 2009; Gupta et al., 2006; Kassotaki, 2022; Lubatkin et al., 2006; O'Reilly & Tushman, 2013; Pertusa-Ortega et al., 2021; Raisch, 2008; Raisch et al., 2009; Tarba et al., 2020). This line of argumentation rests on the assumption that exclusively focusing on exploration activities entails high risks while, on the other hand, exclusively focusing on exploitation activities may lead into a competency trap (Cao et al., 2009; Gupta et al., 2006). Thus, in dynamic environments, firms must therefore pursue exploration and exploitation equally, without compromising one over the other (Lerch et al., 2024). Since exploration and exploitation are two logically distinctive and mutually exclusive means of learning and competing (March, 1991), pursuing organizational ambidexterity comes along with fundamental contradictions, conflicts, tensions or paradoxes (Andriopoulos & Lewis, 2009; Martin et al., 2019).

To handle different logics and contradictions, firms may structurally differentiate yet integrate the two different modes of learning and competition (Fourné et al., 2019; Hansen et al., 2019; Jansen, Tempelaar, et al., 2009; Lawrence & Lorsch, 1967; Ossenbrink et al., 2019; O'Reilly & Tushman, 2013; Raisch, 2008). Differentiation is pursued by establishing independent fields of logic, i.e., exploration and exploitation, while integration is pursued by establishing integration mechanisms such as a shared vision, joint decision-making processes or having internal knowledge flows (Carmeli & Halevi, 2009; Horvat, Dreher, & Som, 2019; Horvat et al., 2024; Lubatkin et al., 2006; Mom et al., 2007). Firm's leaders play a vital role in both differentiation and integration, especially by designing and implementing structural arrangements for separation and integration, or by acting themselves ambidextrous (Carmeli & Halevi, 2009; Hu et al., 2023; Jansen, Tempelaar, et al., 2009; Lubatkin et al., 2006; Tushman, Smith, & Binns, 2011). Koryak et al. (2018, p. 422), for example, conclude that “TMT composition, both in term of heterogeneity and size, matters.” These authors find that TMT heterogeneity, among other factors such as a written vision or R&D-intensity, pushes differentiation forward, instead of being integrative. However, “[o]ne of the most important lessons is that ambidextrous organizations need ambidextrous senior teams and managers—executives who have the ability to understand and be sensitive to the needs of very different kinds of businesses” (O'Reilly & Tushman, 2004, pp. 8–9).

2.2. TMT's innovation orientation and organizational ambidexterity

Given the importance of leaders in pursuing organizational

¹ We use the term “transitional economies” to highlight the specific history and nowadays' institutional setting of formerly planned economies (Round, 2009). Yet, most work in management studies speaks of emerging markets or emerging economies, including transitional economies (Hoskisson, Eden, Lau, & Wright, 2000; Markovic et al., 2021; Pilav – Velic et al., 2024).

ambidexterity, a large body of literature (Buyl et al., 2012; Cao et al., 2010; Dhir et al., 2023; Heavey & Simsek, 2017; Huang et al., 2021; Koryak et al., 2018; Kraiczy et al., 2015; Kwee et al., 2011; Lubatkin et al., 2006; Venugopal et al., 2019, 2020; Wang & Dass, 2017) refers to upper echelon theory (Carpenter et al., 2004; Hambrick et al., 1996; Hambrick & Mason, 1984; Neely et al., 2020) to understand the role of the TMT in organizational ambidexterity. Upper echelon theory argues that a firm's behavior is based on the perception and decisions of the TMT and that these perceptions and decisions are rooted in the characteristics of the TMT (Hambrick & Mason, 1984; Neely et al., 2020). Studies on TMTs in the field of organizational ambidexterity investigate, among other issues, the impact of TMT's diversity (Chen, Miller, & Chen, 2021; Koryak et al., 2018; Li, 2014; Talke et al., 2011), CEO network extensiveness and CEO-TMT-interaction (Cao et al., 2010), distributed cognition in TMTs (Heavey & Simsek, 2017), CEO regulatory focus (Huang et al., 2021), time horizon blending (Chen, Miller, & Chen, 2021), behavioral complexity (Carmeli & Halevi, 2009) or behavioral integration (Carmeli & Halevi, 2009; Lubatkin et al., 2006; Venugopal et al., 2019, 2020) on organizational ambidexterity. Overall, research shows that the TMT and its unique characteristics play a significant role in shaping organizational ambidexterity (Simsek, 2009). Thereby, TMT's innovation orientation is defined as "innovation fostering behaviour of TMT members as a group. It describes the willingness of the TMT to engage in innovative activities" (Kraiczy et al., 2015, p. 4).

2.3. Transitional economies and organizational ambidexterity

Organizational ambidexterity has been widely explored within the framework of developed economies (Batra et al., 2022). However, among other factors, heightened institutional volatility, complexity, and uncertainty, as being prevalent in emerging and transitional economies (Carney et al., 2019; Kostova & Marano, 2018; Round, 2009), introduce distinctive challenges and opportunities for these companies (Dabić et al., 2021; Ferraris et al., 2022; Shankar & Narang, 2020; Sheth, 2011; Stojčić, 2024; Stojčić, 2021; Stojčić et al., 2024; Švarc & Dabić, 2021). This underscores the need for dedicated research into organizational ambidexterity tailored to the specific context of emerging and transitional economies (Batra et al., 2022). These economies deviate from their developed counterparts, presenting substantial variations in both structural and cultural dimensions (Shankar & Narang, 2020).

Structurally, these economies grapple with heterogeneity, sociopolitical governance challenges, resource scarcity, unbranded competition, and deficiencies in infrastructure (Sheth, 2011). Firms in transitional economies, as argued by Radosevic (2017) in respect to Central and Eastern Europe (CEE), often lack essential innovation capabilities, including the skills, knowledge, and experience necessary to identify opportunities, secure resources, and to successfully implement innovation projects (see also Bahl et al., 2021; Cirera & Maloney, 2017; Estrin et al., 2009; Stojčić, 2024; Stojčić, 2021; Stojčić et al., 2020). As a result, these firms are more inclined to utilize existing knowledge rather than generate new insights, leading to innovation patterns that diverge from traditional innovation models. Firms in these economies tend to focus on absorbing imported knowledge and mastering established technologies (Vujanović et al., 2022; Švarc & Dabić, 2021). Consequently, Pilav – Velic et al. (2024) found recently that SMEs in transitional economies focused on exploitative innovation practices, prone to develop evolutionary rather than revolutionary innovations, while Bahl et al. (2021) found evidence that there was a trade-off between internationalization and innovation for entrepreneurial firms in transitional economies, meaning that there was a negative relation between internationalization and innovation. For these economies, Stojčić (2024) concludes "that innovating firms may be prone to cognitive and absorptive limitations that prevent them from targeting the right type of knowledge and using it for their own advantage" (p. 13) and that without institutional support of training innovation activities, firms "on their own struggle to turn asymmetries in their environment into market opportunities due to

systemic failures, policy constraints, and a lack of internal capabilities" (p. 12). Further on, the European Innovation Scoreboard 2024 (European Commission, 2024) summarize that in BiH, R&D expenditure in the public sector, venture capital expenditures and direct and indirect government support of business R&D are relatively low. Overall, the set of challenges associated with operating in these economies is collectively referred to as the liabilities of "emergingness" (Madhok & Keyhani, 2012).

Culturally, emerging and transitional economies stand out due to the pervasive impact of ingrained cultural values and hierarchical structures (Burgess & Steenkamp, 2006). As mentioned above, in transitional economies, socialist values such as collectivism, shared responsibility, employee involvement and limited entrepreneurial mindset shape how businesses operate and how leaders manage. The realm of innovation adds another layer of challenges, encompassing constraints in infrastructure, technological advancement, legal protection, market instability, limited access to affluent consumer bases, and acquiring resources for innovation activities (Cunha et al., 2014).

Accordingly, the successful management of innovation in these economies demands strategies that differ from those in developed economies. While firms in developed economies focus on leveraging their innovation capabilities to drive radical technological advancements, firms in emerging and transitional economies rather need a more fundamental strategy. This involves enhancing the companies' absorptive capacity, building essential innovation and managerial skills, and investing in the necessary workforce development and innovation infrastructure across the entire emerging innovation system (Goñi & Maloney, 2017; Horvat et al., 2024; Stojčić, 2024; Stojčić et al., 2020). Thus, the antecedents and effects of organizational ambidexterity in emerging and transitional economies requires special reconsideration (Batra et al., 2022; Hsu et al., 2013; Hu et al., 2023). Building on this premise, the subsequent section endeavors to formulate a set of testable hypotheses concerning the impact of TMT's innovation orientation on organizational ambidexterity in transitional economies. Precisely, we aim to explore the combined effects of TMT's innovation orientation and organizational innovation culture on organizational ambidexterity and firm performance in transitional economies. In our theorizing, we refer to the balance dimension of organizational ambidexterity (Cao et al., 2009; Kassotaki, 2022).

3. Model and hypotheses

3.1. TMT's innovation orientation, organizational innovation culture and firm performance in transitional economies

Upper echelon theory shows that the TMT's characteristics influence organizational outcomes such as strategic novelty (Hambrick et al., 1996), strategic innovation orientation (Talke et al., 2011), continual growth (Chen et al., 2018), or responsible firm behavior (Chin et al., 2013). Further on, the TMT's characteristics shape organizational structures, processes and attributes by their decision and actions, such as the innovativeness of firms (Talke et al., 2011) or organizational culture and performance (O'Reilly et al., 2014). Kraiczy et al. (2015) demonstrate, for example, that a firm's management innovation orientation is positively correlated with firm growth and that this relationship is mediated through firm innovativeness. Further, TMT's innovation orientation influences the culture of a firm, meaning that the innovative decision-making behavior of the TMT manifests itself in an organizational culture which also fosters innovation (Kraiczy et al., 2015; Talke et al., 2011). We assume that this relation also exists in emerging and transitional economies and we propose the following hypothesis.

H1a. In transitional economies, TMT's innovation orientation has a positive impact on organizational innovation culture.

Wang and Dass (2017) show that top managers' innovativeness is transformed via an exploration orientation into better financial

performance, depending on given firm resources and the industry sector. However, as mentioned above, it is important to consider that firms in transitional economies are influenced by pressures associated with “emergingness” and small firms in these markets must navigate various challenges that require a broad spectrum of knowledge across international, institutional, and technological domains to remain competitive and innovative (Madhok & Keyhani, 2012; Puthusserry et al., 2020; Stojčić et al., 2024). Yet, given limited resources, limited institutional support, and limited innovative capabilities in transitional economies (Bahl et al., 2021; Estrin et al., 2009; Kroll et al., 2017; Radosevic, 2017; Stojčić, 2024), firms must focus on absorbing existing knowledge and mastering established technologies (Pilav – Velic, Jahic, & Krndžija, 2024; Švarc & Dabić, 2021; Vujanović et al., 2022). Therefore, exploitative learning and knowledge is crucial for overcoming structural limitations, adapting to changing market conditions, and effectively integrating into global value chains. Without this comprehensive understanding, firms in transitional economies may struggle to build higher firm performance and to seize growth opportunities. Based on these arguments, we assume the following relation between TMT’s innovation orientation and firm performance, meaning that too much exploration in transitional economies may become dysfunctional for firms.

H1b. In transitional economies, TMT’s innovation orientation has a negative impact on firm performance.

Previous research has demonstrated a positive and significant relationship between organizational culture and firm performance, and thus organizational culture is considered a key factor in explaining why some organizations outperform others in the marketplace (Abdul Rashid et al., 2003; Agostini & Nosella, 2017; Ahmad, 2012; Krašnicka et al., 2018; Ojo, 2010; Rose, Kumar, Abdullah, & Ling, 2008; Zhang et al., 2023). This is especially true for innovation culture, a dimension of organizational culture that influences employees’ decisions and behavior, ultimately contributing to improved firm performance (Duke II & Edet, 2012; Wallach, 1983). Both organizational culture and its integral component – innovation culture – support employees in understanding organizational objectives and contribute to greater (innovation) success. Similar findings have been confirmed in transitional economies, including Central and Eastern European (CEE) region (Krašnicka et al., 2018) and the Southeast European (SEE) region. Specifically, Dabić et al. (2018), based on their study of 253 SMEs in Croatia, found that intellectual capital (including human, structural, and relational capital), along with a robust innovation culture, were vital to a company’s success and were strongly inter-correlated. Therefore, we assume the following hypothesis.

H1c. In transitional economies, organizational innovation culture has a positive impact on firm performance.

Although knowledge of the connection between innovation organizational culture and firm performance is cumulating, research in this area still remains relatively underdeveloped (O’Reilly et al., 2014). De Tienne and Mallette (2012) identified a positive link between innovation-oriented cultures and firm performance, evaluating success through subjective measures like product innovation, business growth, and return on investment. Similarly, Wei et al. (2013) demonstrated that employees’ perceptions of innovation culture were closely tied to how they perceived corporate performance and job satisfaction, highlighting the cultural influence on organizational success. However, contrasting findings were reported by Terziovski (2010) in his study of small and medium-sized enterprises (SMEs) in Australia. This analysis of the relationships among innovative strategies, formal structures, innovation culture, and financial performance did not support the widely accepted belief that a strong innovation culture directly improved performance indicators. One plausible reason for this inconsistency is the complex role of innovation culture, which may act as a mediator or moderator rather than a direct driver of performance. Supporting this reasoning, Krašnicka et al. (2018) discovered that organizational culture partially

mediated the impact of management innovation on firm performance. This suggests that, in specific contexts, a strong organizational culture can amplify the effectiveness of innovation initiatives. These relationships are particularly significant in post-socialist economies, where, under socialist regimes, research and development were predominantly conducted in state-run institutions, leaving private firms with minimal involvement in innovation activities (Dabić et al., 2018). Based on this foundation, we suppose that TMT’s innovation orientation shapes the organizational innovation culture of firms which then impacts firm performance. In other words: we expect to find a decreasing impact of TMT’s innovation orientation on firm performance when taking the impact of organizational innovation culture on firm performance into account. In line with Kraicy et al. (2015), who show that the positive impact of TMT’s innovation orientation on firm growth is mediated through firm’s innovativeness, we assume the following relation.

H1d. In transitional economies, the impact of TMT’s innovation orientation on firm performance is mediated by the organizational innovation culture.

3.2. TMT’s innovation orientation, organizational innovation culture and organizational ambidexterity in transitional economies

Kortmann (2015) shows that the ability of TMTs to make ambidextrous decisions is positively correlated with organizational ambidexterity (innovative ambidexterity). Mom et al. (2007) show that vertical knowledge flows enhance managers exploitation activities but not their exploration activities. Wang and Dass (2017) demonstrate that managers’ innovativeness is positively related to the explore-orientation of firms, while Zang and Li (2017) found that technological capabilities had an inverted U-shaped relationship with innovation ambidexterity. Further, Xie et al. (2022, p. 3) find support for their hypothesis that “[t] here is an inverted U-shaped relationship between collaborative innovation and firms’ innovation performance, such that the impact is initially positive, but becomes more negative as the level of collaborative innovation increases.” Referring to the balance dimension of ambidexterity (Cao et al., 2009; Kassotaki, 2022), we argue that there is an inverted U-shaped relation between TMT’s innovation orientation and organizational ambidexterity, especially in transitional economies in which constraining forces are low and environmental dynamism, uncertainty and volatility are high. Following Zang and Li (2017) and Xie et al. (2022), we therefore state that an increased level of TMT’s innovation orientation leads, after reaching a certain threshold, to an imbalance in the exploration-exploitation relation. This line of argumentation similarly applies to the impact of organizational innovation culture on organizational ambidexterity (BD) and we propose.

H2a. In transitional economies, there is an inverted U-shaped relation between TMT’s innovation orientation and organizational ambidexterity (BD), such that the impact is initially leading to a higher level of balance in ambidexterity, but then ambidexterity becomes imbalanced as TMT’s innovation orientation further increases.

H2b. In transitional economies, there is an inverted U-shaped relation between organizational innovation culture and organizational ambidexterity (BD) such that the impact is initially leading to a higher level of balance in ambidexterity, but then ambidexterity becomes imbalanced as organizational innovation culture further increases.

Furthermore, as argued above, we expect that organizational innovation culture moderates the impact of TMT’s innovation orientation on organizational ambidexterity (BD). Precisely, we expect organizational innovation culture to amplify the impact of TMT’s innovation orientation on organizational ambidexterity (BD). Thus, we state.

H2c. In transitional economies, the inverted U-Shaped relation between TMT’s innovation orientation and organizational ambidexterity (BD) is moderated by organizational innovation culture, so that the inverted U-shaped relationship will be steeper for firms with high

organizational innovation culture.

3.3. Organizational ambidexterity and firm performance in transitional economies

Overall, empirical evidence points to the benefits of organizational ambidexterity in achieving superior firm performance (Gibson & Birkinshaw, 2004; He & Wong, 2004; Junni et al., 2013) and literature argues that this is basically also true for emerging and transitional economies (Batra et al., 2022; Hsu et al., 2013; Hu et al., 2023). Hu et al. (2023), investigating family-managed SMEs in China, find evidence that organizational ambidexterity positively impacts firm performance, although this positive impact is weakened by family members in the TMT of these firms. However, some authors suggest that specialization in exploration or exploitation could be occasionally more beneficial for firms (Benner & Tushman, 2003; Gupta et al., 2006; Siggelkow & Levinthal, 2003). Wenke et al. (2021), for example, show in their meta-study that for SMEs, the relationship of exploration or exploitation with performance is stronger than the relationship of organizational ambidexterity with firm performance. Solís-Molina et al. (2018) demonstrate that exploration or exploitation (specialization) positively impacts firm performance and that combined ambidexterity is only beneficial for firm performance for companies with a high level of absorptive capacity. Gupta et al. (2006) further argue that firms benefit from specialization if they (1) can rely on complementary resources provided externally, (2) when dynamism is either extremely low or extremely high or (3) when the degree of mutual co-specialization in the two sets of resources is low. Consequently, Kafetzopoulos (2021) finds that environmental uncertainty moderates the impact of organizational ambidexterity on business performance.

Since transitional economies are extremely dynamic and volatile, we assume that organizational ambidexterity (BD) is not correlated with firm performance in these settings. SMEs in these economies generally possess lower innovation capacities and competencies, along with limited abilities to engage in continuous learning and adapt to changes in their environment (Bahl et al., 2021; Bodlaj et al., 2020; Burgess & Steenkamp, 2006; Dabić et al., 2021; Estrin et al., 2009; Hervás-Oliver et al., 2021; Stojčić, 2024). They also face significant financial constraints and must navigate the challenges associated with “emergingness”, making it extremely difficult to achieve organizational ambidexterity (BD) that could enhance the overall firm performance. Cao et al. (2009), for example, find evidence that organizational ambidexterity (BD) in resource-constrained contexts is not significantly related to firm performance. Thus, while ambidexterity may be critical for sustained growth and competitiveness, in turbulent and resource-constrained environments, balancing these two distinct and independent processes is challenging. For SMEs in transitional economies, the separation of these processes can help firms to respond more

effectively to market demands and technological changes. However, since exploration requires new capabilities, resources and a certain mindset which SMEs in transitional economies do not (yet) possess, we propose the following hypotheses.

H3a. In transitional economies, organizational ambidexterity (BD) is not correlated with firm performance.

H3b. In transitional economies, an imbalance in organizational ambidexterity (BD) towards exploitation (i.e., specialization in exploitation) is positively correlated with firm performance.

H3c. In transitional economies, an imbalance in organizational ambidexterity (BD) towards exploration (i.e., specialization in exploration) is negatively correlated with firm performance.

Fig. 1 visualizes our model of the impact of TMT’s innovation orientation on organizational ambidexterity (BD) and firm performance via organizational innovation culture.

4. Methodology

4.1. Data base and sample description

To test our hypotheses, we rely on unique company level data of manufacturing SMEs in Bosnia and Herzegovina (BiH), the *BiH SME Innovation Survey 2023*. We chose BiH as a representative example of transitional economies (United Nations, 2024) for several reasons. First, the specific historical, economic, and cultural context of BiH makes it particularly suitable for researching innovation in a complex and highly volatile market setting. BiH is one of the eight countries in the Southeast European (SEE) region, which also includes Slovenia, Croatia, Serbia, Montenegro, North Macedonia, Kosovo, and Albania. This region is known for its complex transition from a centrally planned economy to a market economy, occurring against the historical backdrop of ethnically driven conflicts (Ledeneva & Efendic, 2023). Secondly, BiH is a unique case, serving as a prominent example of a transitional economy characterized by a diverse population with heterogeneous cultural backgrounds. Thirdly, SMEs represent 96 % of the BiH economy. These enterprises encounter persistent challenges, compelled to swiftly adjust and sustain their competitiveness amid harsh conditions marked by limited resources, inadequate legal protection, and market instability which underscore the crucial significance of adopting innovation and suitable ambidextrous strategies in such a demanding business environment (Pilav-Velić & Marjanovic, 2016). Fourthly, one of the key reasons BiH was chosen for the analysis of organizational ambidexterity in SMEs is its classification as an Emerging Innovator in the European Innovation Scoreboard 2024 (European Commission, 2024). Emerging Innovators are economies with innovation performance below 70 % of the EU average. In 2024, BiH’s performance stood at 33.1 % of the EU

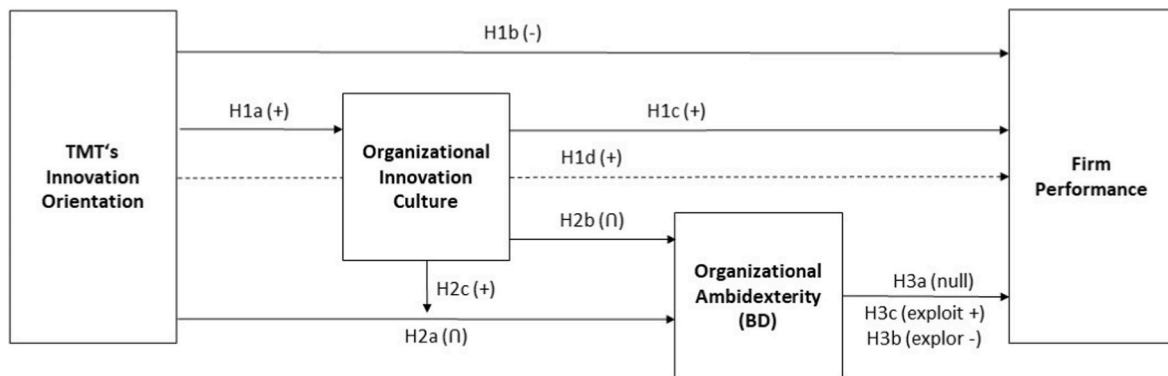


Fig. 1. Model on TMT’s innovation orientation, organizational innovation culture, organizational ambidexterity (BD) and firm performance in transitional economies.

average, which was not only below the Emerging Innovators' average of 48 % but was also growing at a slower rate compared to the EU's overall increase of +10 %. And lastly, we chose to focus on SMEs in BiH because here the impact of the organizational leaders such as the CEO and the TMT seems to be of special importance in understanding organizational ambidexterity (Cao et al., 2010).

The survey focused on firm innovation characteristics. Additionally, a few questions on performance and the competition context as well as basic firm characteristics were included in assessing innovation activities. Before the formal administration of the survey, a rigorous pre-testing phase was undertaken. A round of interviews ($n = 10$) with innovative companies as well as with some experts from academia was conducted (DeVellis, 2016). The *BIH SME Innovation Survey 2023* was then conducted during spring and early summer 2023, organized by the University of Sarajevo, and administered using Computer-Assisted Telephone Interviews (CATI) and Computer-Assisted Web Interviews (CAWI), facilitated by a professional survey company. This survey company also provided the sample frame, so that in total 2181 firms could be addressed (Krndžija, 2024). After thorough checks on data quality and sample inclusion criteria by the service company as well as the research team, the *BIH SME Innovation Survey 2023* data included a total of 247 firms. Our analyses are based on the sub-sample of 192 manufacturing SMEs from this survey. Our data quite well represent the firm size structure of BiH, with small firms as the majority (Keskin et al., 2010, pp. 183–192). Even though the proportion of larger SMEs in our data is most likely higher than in the population, our data scope allows us to take these companies adequately into account. No specific information for industrial SMEs is available for BiH. However, in comparison with the structure of all companies, we can conclude that our data reflects the sectoral distribution, with the metal industry being the largest sector, followed by the food industry. Table 1 provides an overview of these main characteristics of our data. Specifically, 15.7 % of the firms belong to the metal industry, 11 % to the food industry, and another 10.9 % operate in the mechanical, electronic/electrical, and automotive industries. This diverse composition ensures a comprehensive exploration of perspectives within the SME landscape, providing valuable insights into the specific nuances of these sectors.

Moreover, geographical distribution of the survey covers the entire territory of BiH, including both of its entities and the Brčko District. Our sample reflects this geographical distribution proportionately (Krndžija,

2024). Consequently, the survey was conducted in all three official languages—Bosnian, Croatian, and Serbian—offering respondents the option to choose the language in which they preferred to respond. As far as we are aware, this is the first survey of its kind conducted on such a comprehensive scale and with this specific focus on innovation.

4.2. Constructs and measurements

4.2.1. Key impact factors

We regard TMT's innovation orientation (Kraiczay et al., 2015; Wang & Dass, 2017) as a first-order construct that consists of five items. Three of the items were taken from Alheet et al. (2021), while two items were taken from Hoque (2018). For our study, we treat organizational innovation culture as a first-order construct that consists of five items (Kenney & Reedy, 2006). For both constructs, the respondents were asked to indicate how well statements described their firm regarding the respective topic. Both constructs were conceived as a Likert scale (Bernstein, 2005; Jebb et al., 2021; Likert, 1932). We used a seven-point, labelled scale, ranging from 'absolutely not describing' (−3) to 'perfectly describing' (+3). The statements are reported in Table 2, as well as the main descriptive information of the base items. We found these scales to have a Cronbach Alpha of 0.941 resp. 0.955.

4.2.2. Key dependent constructs

There is a vital discussion on how to measure organizational ambidexterity (Cao et al., 2009; He & Wong, 2004; Kassotaki, 2022; Venugopal et al., 2020) and it is commonly suggested to measure exploration and exploitation by making use of separate scales (He & Wong, 2004; Jansen et al., 2006; Jansen, Tempelaar, et al., 2009; Lubatkin et al., 2006). Thereby, organizational ambidexterity is often regarded as the combined result of exploration and exploitation (Jansen, Tempelaar, et al., 2009; Koryak et al., 2018; Lubatkin et al., 2006; Mihalache et al., 2014). However, seeing organizational ambidexterity as a firm's ability to simultaneously pursue exploitation and exploration (Gupta et al., 2006), we follow Cao et al. (2009) and measure the balance dimension of organizational ambidexterity (see also He & Wong, 2004; Venugopal et al., 2020). According to this line of argumentation, "organizational ambidexterity can be described as a midpoint, or an optimal point on a continuum, with exploration and exploitation lying at the two ends" (Kassotaki, 2022, p. 8). It follows that "a higher level of BD, or a closer match in the relative magnitude of exploratory and exploitative activities, contributes to firm performance through more structured control of performance risk" (Cao et al., 2009, p. 783). Hence, organizational ambidexterity (BD) describes the difference between exploitation and exploration, and the lower the difference is, the higher the level of organizational ambidexterity is.

Overall, we use three measures: organizational ambidexterity, the degree of exploitation and the degree of exploration. These measures are based on six statements (see Table 2) for which the respondents were asked to indicate how well these statements described their firm innovation behavior, using a seven-point, labelled scale, ranging from 'absolutely not describing' (−3) to 'perfectly describing' (+3). The six statements were part of a larger item battery (12 items) on firm innovation behavior. The selection of the six statements was based on their capacity to distinguish between explorative and exploitative behavior.

For the *degree of exploitation*, the answers to these questions that classify the statement as more or less not fitting for their company are interpreted as graduated statements on exploitation, i.e., the answers with the values −3 to −1 are recorded as absolute values. All other answers are categorized with the value 0, i.e., not exploitative at all. Consequently, the values range from 0 to 3. In contrast, answers to the six items that rate the statement as more or less fitting for their company are interpreted as statements about the *degree of exploration*. Consequently, the answers with the values 3 to 1 are kept unchanged, while the other neutral or denying responses are categorized as the value 0, indicating being not explorative at all. Thus, the values also range from

Table 1

Overview of our analytic data set (Source: BIH SME Innovation Survey Data 2023; Sarajevo, own calculations).

Constructs	Categories (Code)	Share of Firms
Firm size (# of employees)	10 to 49 employees (1)	62 %
	50 or more employees (0)	38 %
Sector	Metal industry	16 %
	Food industry	11 %
	Mechanical Engineering	5 %
	Electronics/Electrical industry	5 %
	Pharmaceutical/chemical industry	2 %
	Automotive industry	1 %
	Other industrial sector	61 %
Turn-over	Up to 4 million BAM	61 %
	4 - 20 million BAM	31 %
	20 or more million BAM	8 %
Standardization of offer	Customized/order-specific products (1)	42 %
	Standardized products (0)	58 %
Product innovator	Firms that introduced new/importantly modified products since 2019 (1)	69 %
Old products in portfolio	Firms with products that are unchanged since at least 10 years (1)	65 %
Total data set		N = 192

Table 2

Base information for explanatory and dependent constructs (Source: BIH SME Innovation Survey Data 2023; Sarajevo).

Statement	Mean	Median	Mode	Range
TMT's Innovation Orientation (Cronbachs alpha = 0.941, 5 items)				
(1) The management considers innovation to be part of the firm's strategic goals and future ambitions.	1.15	1.00	3	−3 to +3
(2) The management favours a strong emphasis on R&D, technological leadership, and innovations.	0.97	1.00	3	−3 to +3
(3) The management undertakes risky projects to explore opportunities.	0.60	1.00	2 and 3	−3 to +3
(4) The management is involved in new initiatives and innovative programs.	1.07	1.00	3	−3 to +3
(5) The management allocates resources to support the development of new products, services, or processes	1.04	1.00	3	−3 to +3
Organizational Innovation Culture (Cronbachs alpha = 0.955, 5 items)				
(1) The company is open to accepting new ideas and ideas that come from the external environment.	1.85	2.00	3	−2 to +3
(2) Staff within our firm obtains and exchanges new knowledge and skills in fair and collegial ways.	1.70	2.00	3	−2 to +3
(3) The firm shares important information such as success and failure and customer feedback with all relevant staff as part of our internal learning processes.	1.44	2.00	3	−3 to +3
(4) The firm has a flexible organizational structure (for example decentralization, shared decision making, low-moderate use of formal rules ...).	1.46	2.00	3	−3 to +3
(5) The firm encourages staff to think freely, generate ideas, follow-up on ideas, learn experiences, and take risks.	1.63	2.00	3	−3 to +3
Organizational Ambidexterity (Balance Dimension), Degree of Exploitation (Cronbachs alpha = 0.896, 6 items), Degree of Exploration (Cronbachs alpha = 0.932, 6 items)				
(1) The firm frequently tries out new ideas.	1.13	1.00	3	−3 to +3
(2) The firm is a pioneer with introducing new products/services to the market	0.37	1.00	3	−3 to +3
(3) The management seeks out new ways to do things.	1.14	1.00	2	−3 to +3
(4) The firm develops new market segments.	0.90	1.00	3	−3 to +3
(5) The firm is regularly looking for new marketing methods.	0.74	1.00	2 and 3	−3 to +3
(6) The firm seeks new ways to establish customer relationships	1.06	1.00	3	−3 to +3
Firm Performance During Last Year (Cronbachs alpha = 0.967, 3 items; Kendal Tau b = 0.872, p < 0.001 (item 1 and 2); Kendal Tau b = 0.843, p < 0.001 (item 1 and 3), Kendal Tau b = 0.863, p < 0.001 (item 2 and 3))				
(1) return on investment (ROI) compared to the major firm's competitors	−0.83	−1.00	−1	−3 to +3
(2) return on sales (ROS) compared to the major firm's competitors	−0.80	−1.00	−2	−3 to +3
(3) reach of financial goals compared to the major firm's competitors	−0.85	−1.00	−2	−3 to +3

0 to 3. Subsequently, the newly created variables for both constructs are combined using the mean value function. The result is two separate indicators, for the degree of exploitation and the degree of exploration. Thereby, both indicators have a range from 0 to 3, where 3 is the highest degree, which corresponds to the natural understanding of increasing intensity in the respective orientation. We found these scales to have a Cronbach Alpha of 0.896 (exploitation) resp. 0.932 (exploration). Based on these measures, organizational ambidexterity (BD) is calculated as the absolute difference between them. As proposed by [Cao et al. \(2009\)](#), the measure is reversed by subtracting the difference score from 3 so that a higher value indicates greater balance. The organizational ambidexterity (BD) measure has a range between 0 and 3, where 3 indicates the highest level of organizational ambidexterity (BD).

Research on organizational ambidexterity has intensively measured firm performance to understand the effects of organizational ambidexterity ([He & Wong, 2004](#); [Jansen et al., 2006](#); [Lubatkin et al., 2006](#)). In this paper, we focus on the profitability dimension of a firm's performance ([Tarutė & Gatautis, 2014](#)). The measure is operationalized based on three items ([Table 2](#)). For each item, the respondents were asked to assess the profitability of their firm in comparison to their major competitors during the previous year, using a range from 'much worse' (−3) to 'much better' (+3). Although this type of comparative measurement of performance can only provide a rough estimate of financial performance, this approach enables an evaluation across different business contexts. In order not to overestimate individual answers, the information across the items is averaged and a smoothed relative indicator is calculated. This indicator is ranging from 'much worse' (−3) to 'much better' (+3). The inter-correlation between these items is sufficiently high (Kendall Tau b of 0.843, p < 0.001, resp. 0.863, p < 0.001, resp. 0.872 p < 0.001).

4.2.3. Further measures

The variables used to control for established effects include firm size, sector, product standardization, and product innovation. We controlled for firm size because it has been found to influence firm performance. Larger companies have more experience with the introduction of advanced production technologies and higher economies of scale ([Horvat, Kroll, & Jäger, 2019](#)). Firm size was measured as a dichotomous indicator based on the indication of the corresponding firm size class. Together with the standardization of the product offer, the industry can be used to approximate both the competitive context and the production context and capture the average differences in innovation practices ([Jäger et al., 2016](#)). We additionally control for the status as a product innovator, i.e., whether a company has introduced new or majorly improved products since 2019 or not, and for the status as a producer of long-established products, i.e., whether a company has been selling unmodified products for at least 10 years or not. Both indicators are of dichotomous nature. The coding and the average values (shares) of these additional independent indicators can be found in [Table 1](#).

4.2.4. Model description and data analysis

Several OLS regression models are estimated to test our hypotheses. All dependent variables are treated as metric, even if they have a limited range, as they represent latent constructs whose range could be infinite. This is in line with the understanding of the analysis of the Likert scales. We estimate linear regression models. The tables are listed in the order of the dependent variable. The non-linear inverse u-shaped relationships are modelled as the direct influence in combination with the influence of the squared term. Even if the squared term is a direct derivative of the influencing variable, this is not a violation of the requirement for linearly independent columns in the design matrix. The regression was also estimated with the centered variables to check for potential problems of multi-collinearity. No relevant deviations occurred.

5. Results

[Table 3](#) provides the descriptive statistics of our main independent and dependent constructs, i.e., their means (Mean), standard deviations (SD), and correlation between them. The results show that firm performance is negatively related to TMT's innovation orientation and organizational innovation culture, but positively related to organizational ambidexterity (BD). Accordingly, organizational ambidexterity (BD) is negatively related to TMT's innovation orientation and organizational innovation culture. Moreover, even if the degree of exploitation and the degree of exploration are intricately linked with organizational ambidexterity (BD), only the latter is significantly related. Based on these results, the degree of exploitation seems not to be of importance for organizational ambidexterity (BD) in transitional economies. Overall, however, there is no strong correlation between the factors. In addition,

Table 3

Correlation of main variables and descriptive statistics (Source: BIH SME Innovation Survey Data 2023; Sarajevo).

	(1) Organizational ambidexterity (BD)	(2) Degree of exploitation	(3) Degree of exploration	(4) TMT's innovation orientation	(5) Organizational innovation culture	(6) Firm performance
(1) Organizational ambidexterity (BD)	1					
(2) Degree of exploitation	0.036	1				
(3) Degree of exploration	−0.710***	−0.657***	1			
(4) TMT's innovation orientation	−0.323***	−0.708***	0.730***	1		
(5) Organizational innovation culture	−0.243**	−0.675***	0.700***	0.736***	1	
(6) Firm performance	0.202**	0.488***	−0.500***	−0.465***	−0.521***	1
Mean	1.46	0.43	1.32	0.96	1.62	−0.83
Std. Dev.	0.9167	0.6953	1.0097	1.5571	1.3273	1.32149
Min.	0	0	0	−3	−1.6	−3
Max.	3	3	3	3	3	3
N	192	192	192	192	192	192
# of items	6	6	6	5	5	3

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two tailed).

we tested multi-collinearity using the variance inflation factors (VIF) for all predictors in each model. This showed that multi-collinearity was not a major problem. All VIF values were below the threshold of 10.0.

Regressions results are presented in Table 4, Table 5, Tables 6 and 7.

As shown in Table 4, Model 1–2 indicates that there is a statistically significant positive impact of TMT's innovation orientation on organizational innovation culture ($B = 0.615$, $p < 0.001$). When integrating TMT's innovation orientation, adj. R-sq increases from 0.118 to 0.549 ($p < 0.001$). Therefore, **hypothesis H1a is supported**. Analyzing the impact of TMT's innovation orientation and organizational innovation culture on firm performance, Model 2–2 in Table 5 indicates that TMT's innovation orientation has a statistically significant impact on firm performance (Δ adj. R sq = 0.107, $p < 0.001$). As expected, TMT's innovation orientation negatively impacts firm performance in transitional economies ($B = -0.310$, $p < 0.001$). Thus, **H1b is supported**. These findings are echoed by the impact of organizational innovation culture on firm performance where we see a similar pattern. Model 2–3 in Table 5 shows that the impact of organizational innovation culture on firm performance is statistically significant (Δ adj. R sq = 0.166), but negative ($B = -0.436$, $p < 0.001$). Model 2–4 mirrors this result with an estimated coefficient of $B = -0.371$ ($p < 0.001$) for this impact. Thus, **we must reject H1c** since we expected to find a positive impact of innovation orientation culture on firm performance.

As the estimates of Model 2–4 in Table 5 show, the significantly negative impact of TMT's innovation orientation on the firm becomes

statistically insignificant when organizational innovation culture is integrated into the model ($B = -0.081$, $p > 0.1$). Since there is a positive impact of TMT's innovation orientation on organizational innovation culture, as demonstrated in Model 1–2 in Table 4 ($B = 0.615$, $p < 0.001$), the impact of TMT's innovation orientation on firm performance is fully mediated by innovation organization culture (Baron & Kenny, 1986). Integrating TMT's innovation orientation in Model 2–4 leads to no real model improvement (Δ adj. R-sq = 0.001). Taken together, **we find support for H1d**.

H2a states that in transitional economies there is an inverted U-shaped relation between TMT's innovation orientation and organizational ambidexterity (BD). Model 4–2 in Table 7 shows a linear, negative influence of TMT's innovation orientation on organizational ambidexterity (BD) ($B = -0.106$, $p < 0.05$). Considering the curvilinear relationship by integrating TMT's innovation orientation squared, as shown in Model 4-3, the model is improved as the adj. R-sq increases from 0.177 to 0.338. Thereby, the influence of TMT's innovation orientation squared is statistically significant and negative ($B = -0.147$, $p < 0.001$), implying a basic inverted U-shaped relationship. In contrast, the linear influence of TMT's innovation on organizational ambidexterity (BD) disappears almost completely, as the estimated regression coefficient $B = 0.036$ ($p > 0.1$) shows. Therefore, **we find support for H2a**.

H2b states that in transitional economies, there is an inverted U-shaped relation between organizational innovation culture and organizational ambidexterity (BD). The estimations of Model 4–4 in Table 7 indicate that the impact of organizational innovation culture on organizational ambidexterity (BD) is statistically not significant ($B = -0.087$, $p < 0.1$). However, when integrating the curvilinear relation of organizational innovation culture, as seen in Model 4–5 in Table 7, the explanatory power of the model increases as the adj. R-sq rises from 0.164 to 0.299. Both indicators of organizational innovation culture are statistically relevant: The estimation of the linear effect is positive ($B = 0.286$, $p < 0.001$) while the impact of organizational innovation orientation squared is negative ($B = -0.191$, $p < 0.001$). This indicates again an inverted u-shape relation. Therefore, **we find support for H2b**. Model 4–7 in Table 7 shows that the relation between TMT's innovation orientation and organizational ambidexterity (BD) is moderated by organizational innovation culture since the explanatory power of the model is statistically increased by the integration of both interaction terms of organizational innovation culture and TMT's innovation orientation. As the estimation shows, the moderation is linked to the linear effect ($B = -0.156$, $p < 0.01$). This supports **hypothesis H2c**. In Fig. 2, the organizational ambidexterity (BD) is plotted against TMT's innovation orientation while differentiating various levels of organizational innovation culture. The figure is based on the

Table 4

The effect of TMT's innovation orientation on organizational innovation culture, based on subsequent regression estimations.

DV: Organizational Innovation Culture				
	Model 1-1		Model 1-2	
	B	SE	B	SE
Firm size	−0.128	0.188	0.033	0.135
Sector (metal)	0.088	0.253	−0.163	0.182
Sector (food)	0.690*	0.303	−0.084	0.225
Customized products or services	−0.024	0.191	−0.016	0.137
New products	0.784***	0.203	0.441**	0.147
Unchanged products	−0.368+	0.196	0.139	0.145
TMT's innovation orientation			0.615***	0.046
_cons	1.312***	0.268	0.650**	0.198
Model fit				
adj. R-sq	0.118***		0.549***	
Delta adj. R-sq			0.430***	
N	192		192	

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 5

The effect of TMT's innovation orientation and organizational innovation culture on firm performance, based on subsequent regression estimations.

DV: Firm Performance								
	Model 2-1		Model 2-2		Model 2-3		Model 2-4	
	B	SE	B	SE	B	SE	B	SE
Firm size	0.274	0.179	0.193	0.168	0.218	0.160	0.205	0.161
Sector (metal)	−0.686**	0.242	−0.559*	0.227	−0.647**	0.216	−0.620**	0.217
Sector (food)	−0.142	0.290	0.247	0.280	0.159	0.262	0.216	0.268
Customized products or services	0.092	0.183	0.088	0.170	0.082	0.163	0.082	0.163
New products	−0.733***	0.194	−0.560**	0.183	−0.391**	0.180	−0.397*	0.180
Unchanged products	0.758***	0.188	0.502**	0.181	0.598***	0.169	0.554**	0.174
TMT's innovation orientation			−0.310***	0.057			−0.081	0.077
Organizational innovation culture					−0.436***	0.063	−0.371***	0.088
_cons	−0.896**	0.256	−0.563*	0.247	−0.324	0.243	−0.321	0.243
Model fit								
adj. R-sq	0.187		0.294***				0.354***	
Delta adj. R-sq			0.107***		0.353***		0.166***	
N	192		192		192		192	

+p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001.

Table 6

The effect of organizational ambidexterity (BD) and its components on firm performance, based on subsequent regression estimations.

DV: Firm Performance										
	Model 3-1		Model 3-2		Model 3-3		Model 3-4		Model 3-5	
	B	SE	B	SE	B	SE	B	SE	B	SE
Firm size	0.274	0.179	0.271	0.180	0.253	0.171	0.249	0.175	0.247	0.171
Sector (metal)	−0.686**	0.242	−0.674**	0.243	−0.568*	0.233	−0.634**	0.237	−0.570*	0.233
Sector (food)	−0.142	0.290	−0.091	0.298	0.068	0.287	−0.084	0.291	0.044	0.288
Customized products or services	0.092	0.183	0.102	0.184	0.081	0.175	0.076	0.179	0.074	0.175
New products	−0.733***	0.194	−0.718***	0.195	−0.590**	0.188	−0.641***	0.191	−0.581**	0.188
Unchanged products	0.758***	0.188	0.712***	0.198	0.578**	0.191	0.632***	0.194	0.569**	0.191
Organizational ambidexterity (BD)			0.077	0.104	−0.057	0.104	0.242*	0.113	0.030	0.132
Organizational ambidexterity (BD)					0.460***	0.105			0.382**	0.128
X Degree of exploitation										
Organizational ambidexterity (BD)							−0.327**	0.099	−0.127	0.118
X Degree of exploration										
_cons	−0.896***	0.256	−0.999***	0.292	−1.122***	0.280	−0.805**	0.290	−1.026***	0.294
Model fit										
adj. R-sq	0.187***		0.185***		0.258***		0.227***		0.259***	
Delta adj. R-sq			−0.002		0.071***		−0.031**		0.001	
N	192		192		192		192		192	

+p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001.

grouped estimates for the displayed variables. As indicated by this 2D-graph, firms with a very low level of organizational innovation culture would benefit from an increased level of TMT's innovation orientation while in firms with a low to high level of organizational innovation culture, an increased level of TMT's innovation orientation is likely to foster an imbalance in organizational ambidexterity (BD). Thus, the level of organizational innovation culture moderates the relation between TMT's innovation orientation and organizational ambidexterity (BD) so that the inverted U-shaped relationship is steeper for firms with high organizational innovation cultures. **We therefore find support for hypothesis H2c.**

Hypothesis H3a, H3b, and H3c relate to the impact of organizational ambidexterity (BD) on firm performance in transitional economies. As seen in Model 3–2 in Table 6, organizational ambidexterity (BD) has no significant impact on firm performance ($B = 0.077$, $p > 0.1$, no increase in adj. R-sq). Thus, **we find support for H3a** since we stated that there was no correlation between organizational ambidexterity (BD) and firm performance. As shown in Model 3–3 in Table 6, however, an imbalance in ambidexterity towards exploitation (i.e., exploitation exceeds exploration) positively impacts firm performance ($B = 0.460$, $p < 0.001$). In contrast, as Model 3–4 in Table 6 shows, imbalanced organizational

ambidexterity (BD) towards exploration negatively impacts firm performance ($B = -0.327$, $p < 0.01$). Taking both imbalances into account, it becomes visible, as Model 3–5 in Table 6 shows, that the positive correlation of an imbalance towards exploitation provides explanatory contribution. Firm performance in transitional economies benefit from this imbalance. **In conclusion, the results suggest support for H3b and for H3c.**

6. Discussion

Tackling the complex relation between TMT's innovation orientation and organizational innovation culture and their joint impact on the balance dimension of organizational ambidexterity and firm performance, we offer a holistic understanding of the antecedents and effects of organizational ambidexterity (BD) in transitional economies. Intriguingly, our findings challenge the notion that organizational ambidexterity is beneficial for firms in emerging markets (Batra et al., 2022; Hsu et al., 2013; Hu et al., 2023). Overall, our results confirm the findings provided by Venugopal et al. (2020) who demonstrate that the balance dimension of organizational ambidexterity is not correlated with firm performance while the combined dimension of organizational

Table 7
The effect TMT's innovation orientation and organizational innovation culture on organizational ambidexterity (BD), based on subsequent regression estimations.

DV: Organizational Ambidexterity (BD)

	Model 4-1		Model 4-2		Model 4-3		Model 4-4		Model 4-5		Model 4-6		Model 4-7	
	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE	B	SE
Firm size	0.038	0.127	0.010	0.126	-0.024	0.113	0.027	0.126	0.106	0.116	0.034	0.111	0.058	0.109
Sector (metal)	-0.149	0.171	-0.105	0.170	-0.124	0.152	-0.141	0.170	-0.170	0.156	-0.139	0.148	-0.128	0.145
Sector (food)	-0.660**	0.205	-0.526*	0.209	-0.369+	0.189	-0.600**	0.207	-0.469*	0.191	-0.317+	0.184	-0.266	0.183
Customized products or services	-0.128	0.129	-0.129	0.128	-0.171	0.115	-0.130	0.129	-0.055	0.118	-0.113	0.113	-0.093	0.111
New products	-0.195	0.137	-0.136	0.137	-0.070	0.124	-0.127	0.142	-0.099	0.130	-0.045	0.123	-0.068	0.121
Unchanged products	0.599***	0.133	0.511***	0.136	0.365**	0.124	0.567***	0.133	0.515***	0.122	0.368*	0.120	0.325**	0.119
TMT's innovation orientation			-0.106*	0.043	0.036	0.044					0.012	0.064	0.190+	0.102
TMT's innovation orientation - squared					-0.147***	0.022					-0.107***	0.025	-0.068+	0.041
Organizational innovation culture							-0.087+	0.049	0.286***	0.077	0.193+	0.102	-0.039	0.127
Organizational innovation culture - squared									-0.191***	0.032	-0.121***	0.035	-0.030	0.049
Organizational innovation culture X TMT's innovation orientation													-0.156**	0.055
Organizational innovation culture X TMT's innovation orientation squared													0.011	0.016
_cons	1.335***	0.181	1.450***	0.185	1.876***	0.177	1.449***	0.192	1.604***	0.177	1.899***	0.180	1.995***	0.189
Model fit														
adj. R-sq	0.155***		0.177***		0.338***		0.164***		0.299***		0.378***		0.401***	
Delta adj. R-sq	0.023*		0.023*		0.161***		-0.174+		0.135***		0.082**		0.023*	
N	192		192		192		192		192		192		192	

+p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001.

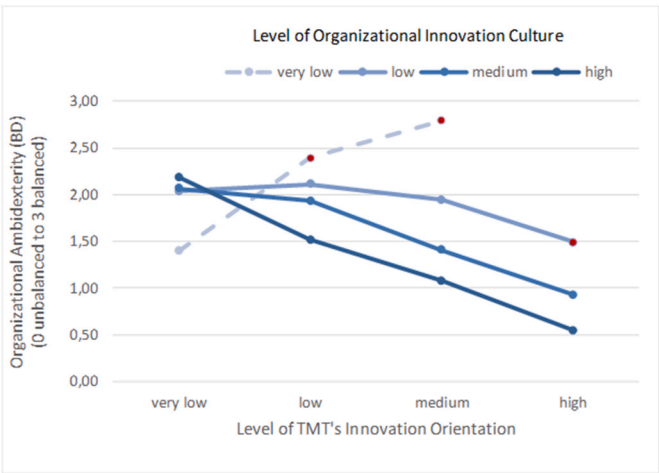


Fig. 2. The moderating effect of organizational innovation culture on the relationship between TMT's innovation orientation and organizational ambidexterity (BD) (Note: Red marked dots are of theoretical nature, as empirically only 1 or 2 companies belong to this class). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

ambidexterity positively impacts firm performance (see also [Cao et al., 2009](#)). We further show that in transitional economies, TMT's innovation orientation is negatively correlated with superior firm performance. We see the same for organizational innovation culture. Although we have to reject hypothesis [H1c](#), our findings are consistent with research on innovation behavior in transitional economies, stating that due to their institutional framework and the lack of innovation capabilities, firms tend (and are recommended) to rely on exploitative and incremental innovation to prosper ([Bahl et al., 2021](#); [Cirera & Maloney, 2017](#); [Dabić et al., 2021](#); [Estrin et al., 2009](#); [Pilav – Velic, Jahic, & Krndzija, 2024](#); [Radosevic, 2017](#); [Stojcic, 2024](#); [Vujanović et al., 2022](#); [Švarc & Dabić, 2021](#)).

This picture is echoed when looking at the effects of the components of organizational ambidexterity on firm performance. We find that an imbalance in organizational ambidexterity towards exploration negatively impacts firm performance, while an imbalance towards exploitation positively impacts firm performance. This consistently underlines our above-mentioned argument, that too much of exploratory innovation orientation may be dysfunctional for the performance of firms in transitional economies. A similar conclusion was reached by [Pilav – Velic et al. \(2024\)](#) who highlighted the specifics of exploration and exploitation in SMEs in BiH with a particular focus on firm resilience. Without considering the balance dimension of organizational ambidexterity and its role in achieving firm performance, this study found that in companies with a higher level of firm resilience, exploitative innovation influenced innovative performance.

When looking at the antecedents, we make two major contributions in understanding organizational ambidexterity (BD). First, corresponding with [Zang and Li \(2017\)](#) and [Xie et al. \(2022\)](#), we see that the relation between TMT's innovation orientation and organizational ambidexterity (BD) is inverted U-shaped, and the same applies for the relation between organizational innovation culture and organizational ambidexterity (BD). Therefore, only for firms with low levels of TMT's innovation orientation, it may be beneficial to strengthen their TMT's innovation orientation when they strive for organizational ambidexterity (BD). Second, strengthening a firm's TMT's innovation orientation is a tightrope act because it comes not only with lower firm performance (mediated through organizational innovation culture) but also leads to an imbalance in organizational ambidexterity (BD). Even more, the relation between TMT's innovation orientation and organizational ambidexterity (BD) is facilitated by innovation organization culture

which itself is affected by TMT's innovation orientation. Firms with a relatively low level of TMT's innovation orientation therefore may benefit from this joint effect while firms with higher levels of TMT's innovation orientation may find themselves having an (overly) extreme focus on exploration, with an imbalance in organizational ambidexterity and low firm performance.

Overall, our results show that in transitional economies, exploration seems to be a risky endeavor for SMEs. For these economies, we cannot confirm that organizational ambidexterity (BD) is associated with greater firm performance (Cao et al., 2009; Venugopal et al., 2020). However, it is important to note that our findings are based on the observation of SMEs. Therefore, we cannot exclude the possibility that large firms in transitional economies, in contrast to SMEs, would benefit from organizational ambidexterity (BD). However, our findings suggest that in transitional economies, focusing on exploitation is beneficial for firms (Pilav – Velic, Jahic, & Krndzija, 2024; Pilav-Velić & Marjanovic, 2016). Thereby, our study deviates from Cao et al. (2009) who find for China-based SMEs, operating in high technology sectors, a positive relation between exploration and firm performance but no significant relation between exploitation and firm performance. In our case, the opposite applies, which may be due to a less innovation supportive intuitional setting (Radosevic, 2017; Stojcic, 2024).

7. Implications

Investigating the organizational ambidexterity-performance-linkage, Junni et al. (2013, p. 308) “found that OA is less important in manufacturing than in service and high-technology sectors.” These findings tend to correspond with our observations, and those of others. For example, a study conducted among Jordan industrial companies, Al-Khatib and Al-Ghanem (2022) found that exploration (radical innovation) and exploitation (incremental innovation) accounted for more than 60 % of the variance in competitive advantage. These authors conclude “that the relationship between radical innovation and competitive advantage is modified through the high-tech industries. Meanwhile, the relationship between incremental innovation and competitive advantage is modified through the low-tech industries” (Al-Khatib & Al-Ghanem, 2022, p. 344). However, it is important to note that Junni et al. (2013) argue that the lower importance of organizational ambidexterity in manufacturing is due the little dynamism in this industry sector. Our study suggests the opposite: In extremely dynamic and volatile environments, such as emerging and transitional economies, organizational ambidexterity may become dysfunctional (Gupta et al., 2006; Luger et al., 2018; Siggelkow & Levinthal, 2003). Overall, external contingencies play a key role in understanding organizational ambidexterity (Cao et al., 2009; Luger et al., 2018), especially when it comes to transitional economies. Therefore, one important implication of this study is that research on organizational ambidexterity needs to account for environmental contingencies in a more comprehensive and detailed way, especially when it comes to understand organizational ambidexterity in transitional economies.

Our findings also have significant practical implications for managers operating in transitional economies. In contrast to the commonly accepted notion that organizational ambidexterity universally brings benefits, managers in these economies should approach this strategy with careful consideration. Our study emphasizes the potential disadvantages of both striving for balance organizational ambidexterity (i.e., simultaneously pursuing exploitation and exploration) and striving for (too much) exploration. In these economies, it seems for firms better to react upon market demands and act defensively instead of aiming to create radically new innovations. One central insight for managers revolves around the necessity of embracing a more conservative strategic decision-making approach closely aligned with market demands. This involves giving precedence to an incremental innovation stance rather than pursuing radical innovation at any price. Furthermore, actively managing a firm's organizational innovation culture becomes crucial,

ensuring that it aligns harmoniously with the organization's strategic goals. Finally, managers should adopt a tailored strategic approach, considering industry specifics, market dynamics, and the organization's existing level of ambidexterity. This nuanced approach will empower managers to navigate the complexities and to deal with uncertainties of transitional economies effectively.

8. Limitations and future research

One important limitation of our study relates to the problem of generalizability. Our findings derive from observations conducted in BiH with its unique history and its unique economic, social and cultural characteristics. Albeit BiH represents a transitional economy (Round, 2009; United Nations, 2024) and is therefore fully suited to investigate the antecedents and effects of organizational ambidexterity in transitional economies, we need to be aware of drawing premature conclusions about other countries (see also Cao et al., 2010). Also, the study is based on a data set that was for the first time conducted in BiH on such a comprehensive scale and with a specific focus on innovation. This makes it difficult to definitively assess the quality of the data. We hope that further empirical work for BiH will soon be available to deepen this path of quantitative research. Another limitation is the measurement of organizational ambidexterity and firm performance in our study. While most literature refers to questionnaires provided by Jansen et al. (2006) or Lubatkin et al. (2006), we used a diverse set of items to measure the balance dimension of organizational ambidexterity. This comes with problems of comparability, and it follows that relating our findings to the findings of others needs to be carefully done, especially when compared to studies referring to the combined dimension of organizational ambidexterity (Cao et al., 2009). Further on, we measured the balance dimension of organizational ambidexterity which captured performance effects worse than the combined dimension of organizational ambidexterity (Junni et al., 2013; Zang & Li, 2017). However, it is important to note that our findings are internally consistent. For example, our constructs of TMT's innovation orientation, organizational innovation culture and the degree of exploration are independently measured, but all of them negatively impact firm performance. Thus, our finding that radical innovation activities in transitional economies tend to be not beneficial for firms, as argued above, seems to be solidly grounded. And lastly, the measurement of firm performance, based on three items and reflecting success over the period of one year, may only partly inform about the full effects of organizational ambidexterity on firm performance, especially when taking the long-term benefits of explorative innovation strategies into account.

While organizational ambidexterity in transitional economies has received limited attention from researchers, a recent study by Pilav – Velic et al. (2024) shed light on the unique dynamics of ambidexterity in SMEs within this context, with a particular emphasis on firm resilience. The study revealed that in firms with higher resilience, exploitative innovation positively impacts innovative performance, but without accounting for the role of the balance dimension of organizational ambidexterity in driving performance outcomes. The authors highlight the need for further research, especially since the study was conducted during the COVID-19 pandemic—a period when firm resilience was critical for survival, and most companies prioritized exploitative over exploratory innovation practices. This creates an opportunity for more comprehensive research on how organizational ambidexterity influences the performance of SMEs in transitional economies during (relatively) stable periods. Future studies may also investigate the factors that support the shift from exploitative to exploratory innovation, including organizational culture, leadership styles, access to financing, and specific market dynamics, among other factors. Such research could offer deep insights into how ambidexterity fosters sustainable growth and innovative performance in both turbulent and stable economic conditions. Considering that the cultural factors of a society influence both the organizational culture of companies and the innovation

orientation of the TMT, it is necessary for future research to investigate the impact of these factors in greater detail. In the past, BiH's post-socialist economy fostered a collectivist mindset in SMEs, promoting team cohesion but discouraging individual initiative and risk-taking, which hindered innovation. Hierarchical structures and centralized decision-making limited employee involvement and slowed the adoption of new ideas. Economic instability led SME's TMTs to prioritize short-term survival over long-term innovation, while traditional gender roles and bureaucratic challenges further restricted access to resources and external partnerships. These factors collectively resulted in cautious innovation strategies and limited transformative growth. Therefore, future research agenda should include the examination of the role of social and cultural values on TMT's innovation orientation and the organizational innovation culture. This may also provide the opportunity to know more about how the TMT (and other actors) actually handle the contradictory requirements of exploration and exploitation in transitional economies in their daily operations, for example by applying a paradoxical perspective (Andriopoulos & Lewis, 2009, 2010; Lewis et al., 2014; Smith & Tushman, 2005). Additionally, our findings are based on SMEs in transitional economies. As mentioned above, it would be interesting to see if our findings would apply as well to large firms in transitional economies. We encourage research to enlarge the scope towards larger firms when it comes to understand organizational ambidexterity in transitional economies.

In summary, further research needs to be done on the contingencies of organizational ambidexterity. Investigating transitional economies, in which environmental contingencies such as dynamism, volatility, uncertainty or culture are unique and more prevailing than in developed economies, seems to be a promising research path. We therefore suggest investigating emerging and transitional economies and the complex relation among drivers, moderators, and effects of organizational ambidexterity. Our findings indicate that organizational ambidexterity in transitional economies is different from that in developed markets. This calls for a separate research agenda on organizational ambidexterity in emerging and transitional economies (Batra et al., 2022; Hsu et al., 2013).

CRediT authorship contribution statement

Alexander Martin: Writing – original draft, Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Djerdj Horvat:** Writing – original draft, Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Angela Jäger:** Writing – original draft, Writing – review & editing, Methodology, Formal analysis, Data curation. **Amila Pilav-Velić:** Writing – original draft, Writing – review & editing, Methodology, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Djerdj Horvat, Angela Jäger, Amila Pilav-Velić reports financial support was provided by Germany Federal Ministry of Education and Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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